

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Bachelor of Technology (Civil and infrastructure Engineering) SEMESTER - 5 Winter 2025 (Supply.)

Course :Bachelor of Technology (Civil and infrastructure Engineering) Branch : Engineering and Technology

Semester : SEMESTER - 5

Subject Code & Name: BTCIPE506E - GEOMATICS

Time : 3 Hours]

[Total Marks : 60

Instructions to the Students:

1. Each question carries 12 marks.
 2. Question No. 1 will be compulsory and include objective-type questions.
 3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6
 4. Use of non-programmable scientific calculators is allowed.
 5. Assume suitable data wherever necessary and mention it clearly.
-

- Q1. Objective type questions. (Compulsory Question) 12
1. What is the main principle of tachometry?
 - A) Measuring angles only
 - B) Measuring distances indirectly using angular observations
 - C) Using satellite signals
 - D) Direct tape measurement
 2. In the stadia method of tachometry, the line of sight should ideally be:
 - A) Vertical
 - B) Slightly inclined
 - C) Horizontal or with small inclination
 - D) Perpendicular to staff
 3. The main advantage of Total Station over conventional EDM + Theodolite is:
 - A) Cheaper instrument
 - B) Requires large manpower
 - C) Fully automated angle and distance measurement
 - D) No need for prism
 4. In triangulation, the accuracy primarily depends on:
 - A) Length of base line
 - B) Number of towers
 - C) Type of compass
 - D) Scale of plotting
 5. A satellite station is used when:
 - A) Main station is inside a forest
 - B) Main station point is inaccessible or obstructed
 - C) Base line is very small
 - D) Triangulation is of first order
 6. A subtense bar is primarily used for measuring:
 - A) Heights
 - B) Horizontal angles
 - C) Distances by observing a fixed angle subtended
 - D) Zenith angles

- 7 The latitude of a place can be determined by observing:
 - A) Altitude of sun at noon
 - B) Azimuth of sun at sunrise
 - C) Declination of moon
 - D) Angle of Polaris with vertical
- 8 True bearing can be obtained by astronomical observation because:
 - A) Magnetic field is constant
 - B) Sun's position is fixed
 - C) Pole star lies very close to the north celestial pole
 - D) Instrument error is zero
- 9 Relief displacement in a vertical photograph is:
 - A) Zero everywhere
 - B) Radial from the principal point
 - C) Tangential to nadir point
 - D) Highest at photo center
- 10 Stereoscopy in photogrammetry enables:
 - A) Image enlargement
 - B) Measurement of coordinates
 - C) Perception of depth (3D view)
 - D) Correction of tilt
- 11 GPS determines position using:
 - A) Triangulation based on angles
 - B) Trilateration using satellite ranges
 - C) Height of satellites
 - D) Aerial photographs
- 12 In GIS, spatial data refers to:
 - A) Population statistics
 - B) Geometry and location of objects
 - C) Textual descriptions
 - D) Tabulated census data only

Q2. Solve the following.

- A) Explain the principle of tachometry. Describe the stadia method and derive the general distance formula. 6
- B) What is an Electronic Distance Measurement (EDM) instrument and what are its principles of operation? 6

Q3. Solve the following.

- A) Explain the principle of triangulation and describe the different orders of triangulation systems. 6
- B) What is spherical excess, and how is it accounted for in triangulation surveys? 6

Q4. Solve Any Two of the following.

- A) Explain how latitude can be determined by astronomical observations on the Sun. 6
- B) Describe the purpose and design considerations for transition curves. 6
- C) Define the terms: 6
 1. Zenith & Nadir
 2. Azimuth & Declination
 3. Equinoctial Points

- Q5. Solve Any Two of the following.
- A) Define photogrammetry and explain its main types. 6
 - B) What is relief displacement and how is it used in photogrammetry? 6
 - C) What is stereoscopy? Explain how stereo pairs are used for 3D interpretation. 6
- Q6. Solve Any Two of the following.
- A) Explain the basic principles of remote sensing and describe the components of a remote sensing system. 6
 - B) Explain the advantages and limitations of remote sensing for civil engineering applications. 6
 - C) Write a detailed note on the applications of remote sensing and GIS in surveying and mapping. 6

*** End ***